SakhiShield

Safe access. Trusted support. Greater empowerment.

Milestone 2: Project Design Description | Team EmpowerHer | SDG 5

1. Project Overview

SakhiShield is a digital safety and empowerment platform that helps women and girls send emergency SOS alerts to trusted contacts and access emergency helplines before an unsafe situation escalates. Regular safety apps can expose personal location data through continuous tracking or cause confusion when buttons are pressed by mistake.

The student prototype is a web page that demonstrates the main workflow: a user creates an SOS alert (with a 10-second cancel option for accidental triggers and consent-based location sharing), a trusted contact views and acknowledges it, and the alert moves through the statuses **Active, Acknowledged, Assisted, and Safe**. A summary page records the resolved alerts and lists emergency helplines (112, 1091, 181). The platform only coordinates information. It cannot replace police response or physical rescue services.

2. System Architecture

The prototype runs entirely in the web browser. It has three logical parts: `index.html` holds the structure (sections, form, buttons, table), `style.css` holds the design, and `script.js` holds the logic that runs the 10-second cancel timer, adds SOS alerts, and updates their status. The data is a list of alerts stored inside `script.js`, so it uses sample data and resets when the page is refreshed.

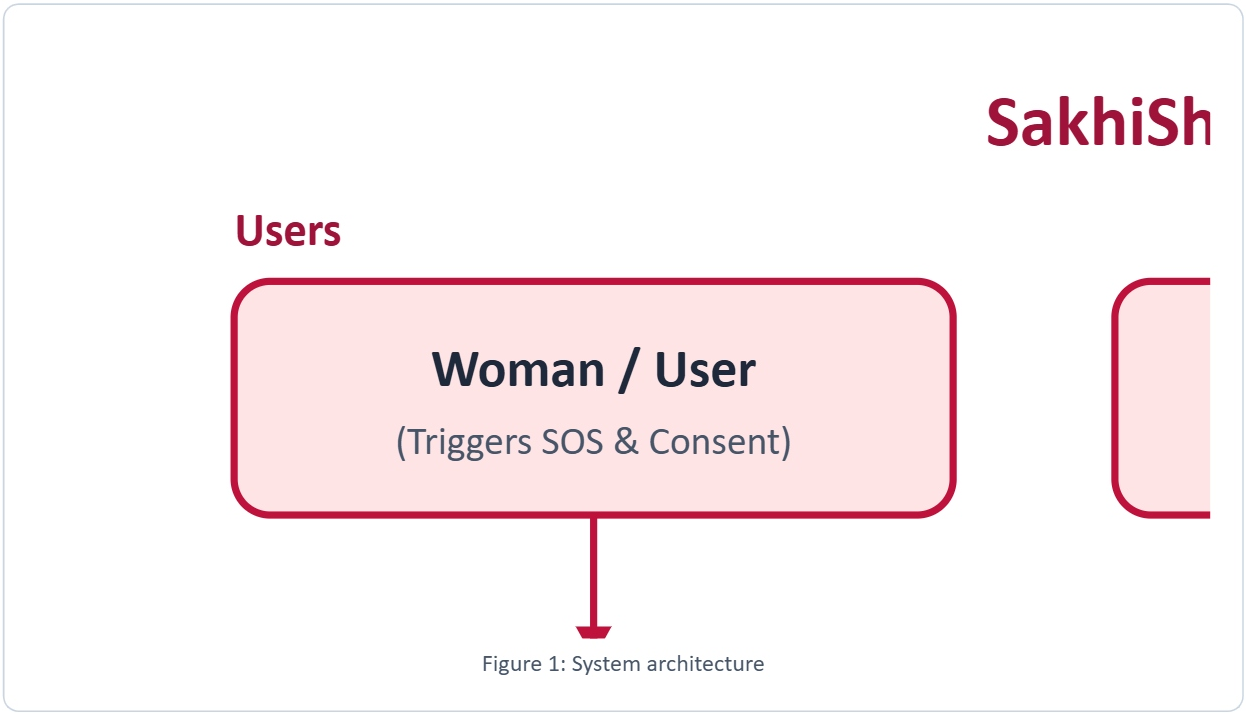


Figure 1: System architecture

3. Workflow

Every SOS alert moves through four statuses in a fixed order. A user adds the SOS alert, which starts as **Active**

(unless cancelled during the 10-second window). A trusted contact acknowledges it, and then the response is marked **Assisted** and finally **Safe**. Only Safe alerts are counted in the resolved total.

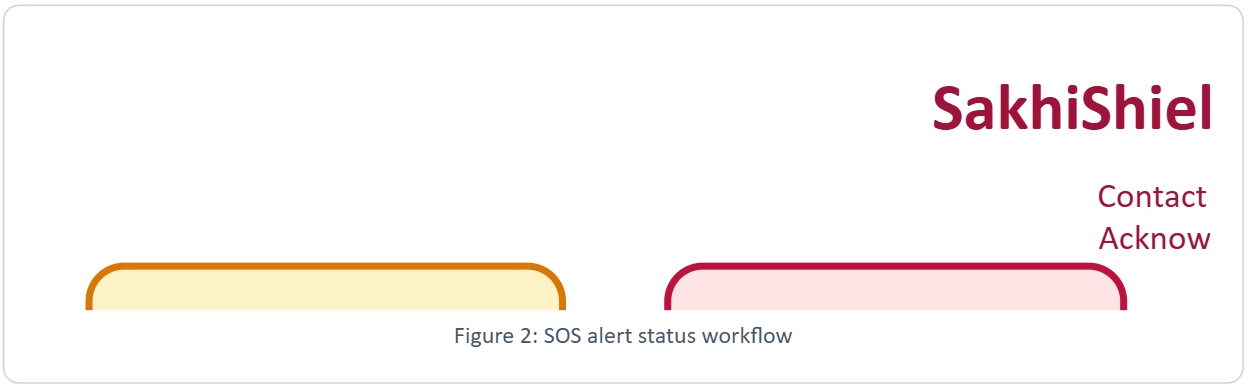


Figure 2: SOS alert status workflow

4. Data Flow

The diagram shows how information moves from the user to the summary page.

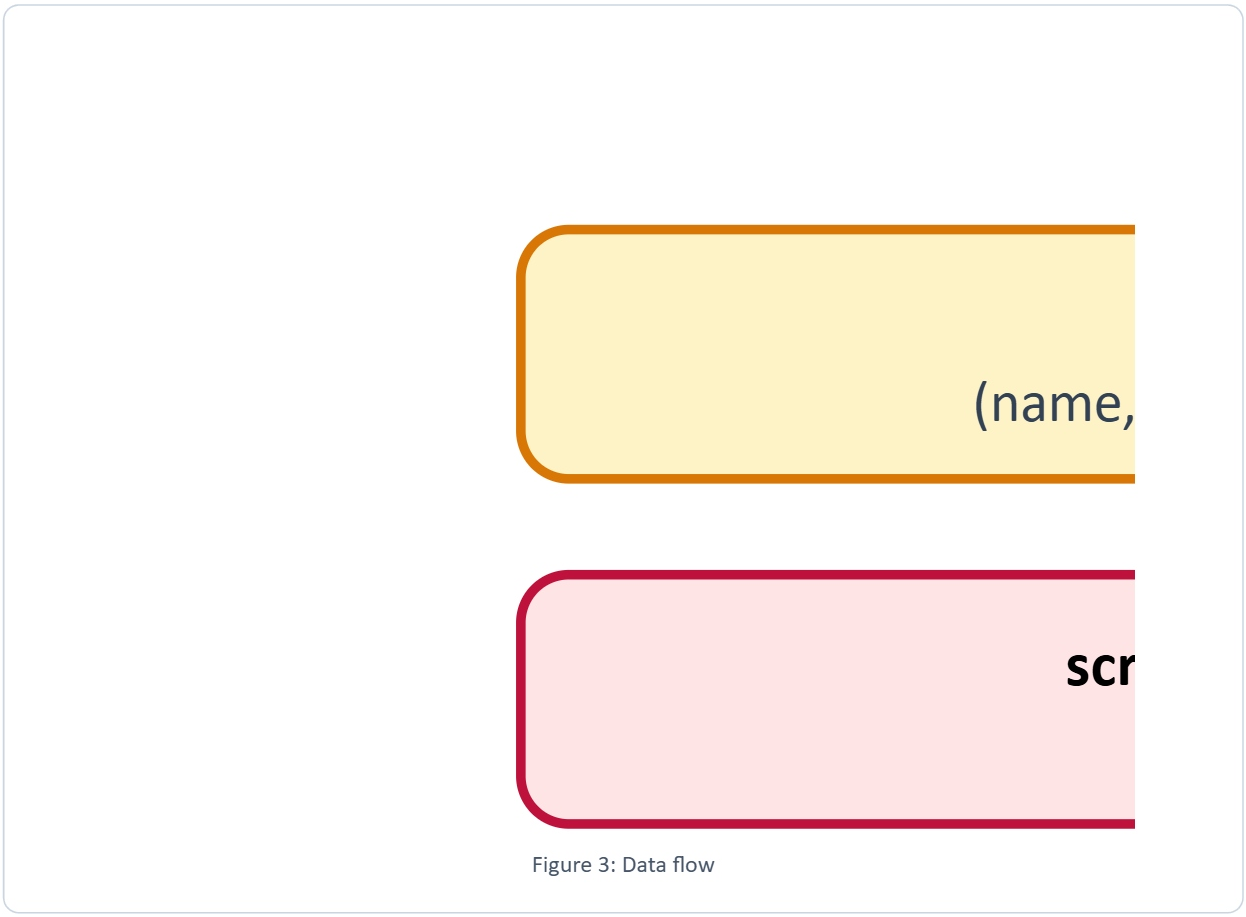


Figure 3: Data flow

5. Data Fields

Each SOS alert stores the following information.

Field	Meaning	Example
`id`	Unique number of the SOS alert	3
`name`	User name	Ananya Sharma (sample)
`location`	Current location shared	Near Metro Gate 2
`type`	Emergency category	Unsafe Cab Ride

`consent`	Consent-based location mode	SOS-Only Encrypted Share
`notes`	Additional safety notes	Battery at 25%
`status`	Active, Acknowledged, Assisted or Safe	Acknowledged

6. User Interface Wireframes

The interface has four screens, which are shown in the menu at the top: Home, Send SOS, Trusted Contacts, and Status & Helplines.

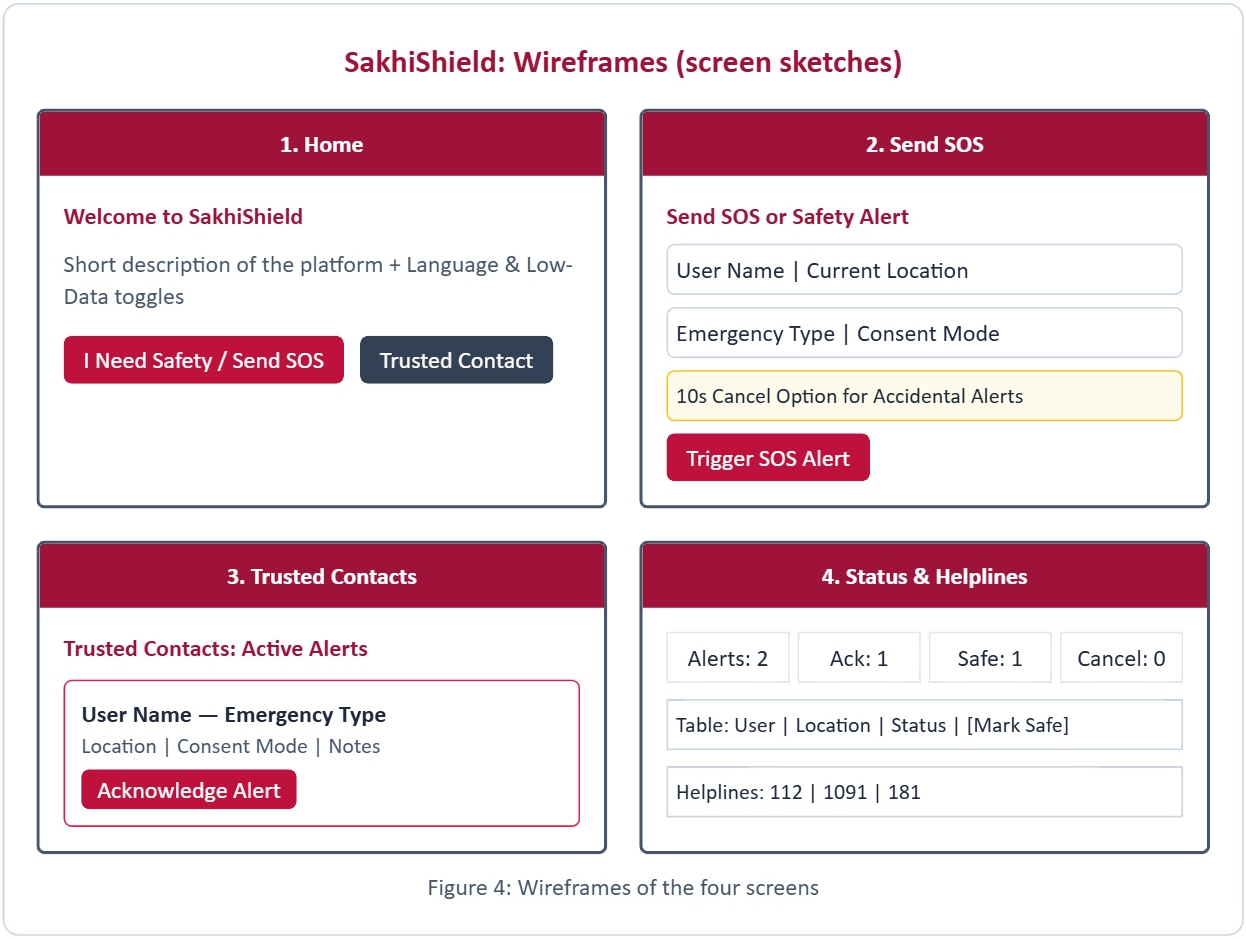


Figure 4: Wireframes of the four screens

7. Features Built and Features Planned

Built in the Milestone 2 prototype:

- SOS form to create an emergency alert with a 10-second cancel option for accidental triggers
- Trusted Contacts view with Active alerts and an Acknowledge button
- Status updates: Acknowledged, Assisted, Safe
- Summary of alerts created, acknowledged, safe, and cancelled false alerts (sample data), plus helplines (112, 1091, 181)

Planned for later and not built yet:

- Trusted contact verification and user login
- Campus security coordinator role
- A real database so data is saved after refresh
- Real telecom SMS gateway and live hardware GPS integration

8. Responsible Use

5. Responsible Use

All data shown in the prototype is sample data created for testing. No real impact figures are claimed. Real use would require verified contacts, live telecom permissions, and emergency response handled by trained people.